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Consistency of maximum likelihood estimators for the regime-switching GARCH model

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The regime-switching GARCH (generalized autoregressive conditionally heteroscedastic) model incorporates the idea of Markov switching into the more restrictive GARCH model, which significantly extends the GARCH model. However, the statistical inference for such an extended model is rather difficult because observations at any time point then depend on the whole regime path and the likelihood becomes intractable quickly as the length of observations increases. In this paper, by transforming it into an infinite order ARCH model, we obtain the possibility of writing a likelihood which can be handled directly and the consistency of the maximum likelihood estimators is proved. Simulation studies to illustrate the consistency and asymptotic normality of the estimators (for both Gaussian and non-Gaussian innovations) and a model specification problem are presented.

Keywords: GARCH model; regime switching; MLE; consistency; asymptotic normality; model specification

AMS 2000 Subject Classification: 62F12; 62M10; 62P20

1. Introduction

The regime-switching GARCH model studied by Cai [1], Hamilton and Susmel [2], Gray [3] and Francq *et al.* [4] among others combined the seminal work of Engle [5] for autoregressive conditional heteroscedastic (ARCH) model (which was later generalized to GARCH by Bollerslev [6]), and the idea of Hamilton [7] for Markov switching models. The main idea is that the set of parameters in GARCH model is determined by some unobservable regimes, where the switch of regimes is usually assumed to be governed by a Markov chain. Empirical results show that it gives more reasonable volatility fit than the ordinary (single-regime) GARCH model (see [3], among others). Also, the strong persistence of the ordinary GARCH processes frequently observed (e.g., refs. [1,8]) can possibly be explained by changes of regimes instead of the integrated GARCH (IGARCH) models. We further note that the regime-switching GARCH model can also be considered as an extension of the hidden Markov model (HMM), which is popular in various fields such as engineering, genetic biology and statistics. For HMM, readers are referred to the monograph by MacDonald and Zucchini [9].

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The statistical inference for the regime-switching GARCH model, however, is rather difficult mainly because observations at every time point depend on all previous regimes due to the autoregressive structure of GARCH equation, which causes an intractable likelihood quickly since summands in the likelihood grow exponentially with the sample size. Gray [3] proposed a reduced regime-switching model in which the previous regime is integrated out at every step and hence the regime path dependence problem is overcome. Klaassen [10] modified it by making use of information of one more observation than Gray's model and showed superiority in prediction.

Compared with the large amount of empirical literature, there are only a few theoretical results. Cai [1] gave a sufficient and necessary condition for the stationarity of the regime-switching ARCH model. This result was extended to the GARCH case by Francq *et al.* [4], where they also proved the consistency of the maximum likelihood estimator (MLE) in the ARCH case. Xie and Yu [11] obtained the consistency of MLE for the reduced regime-switching GARCH model generalized from Gray [3]. They also illustrated the asymptotic behaviour of MLE through simulation study.

In this paper, we will consider the estimation of parameters of a general regime-switching GARCH model. Motivated from Berkes *et al.* [12], we will write the GARCH as an infinite order ARCH model. Therefore, we overcome the path-dependence problem and obtain the likelihood function that can be handled directly. The consistency (Gaussian) of MLE is proved using similar technique to Francq *et al.* [4]. Simulation studies to illustrate the consistency and asymptotic normality of MLE, for both Gaussian and non-Gaussian innovations are presented. In addition, in a model specification problem, we are interested in the persistence if we wrongly use an ordinary GARCH model to fit data generated from regime-switching models. The result is interesting.

We give the model and the likelihood in Section 2. The consistency is proved in Section 3. In Section 4, we give simulation studies. Section 5 includes some final remarks.

2. The model and likelihood

The general regime-switching GARCH(p, q) process $\{Y_t\}_{t\in\mathbb{Z}}$ satisfies

$$\begin{aligned} Y_t &= (h_t)^{1/2}\eta_t, \\ h_t &= \omega(R_t) + \sum_{i=1}^{q} \alpha_i(R_t) Y_{t-i}^2 + \sum_{j=1}^{p} \beta_j(R_t) h_{t-j}, \end{aligned} \tag{1}$$

where $\{\eta_t\}$ is a sequence of independent and identically distributed (i.i.d.) random variables with zero mean and unit variance. h_t is the conditional variance of Y_t given information up to time $t-1$. $\{R_t\}$ is a Markov chain with finite state space $\mathbf{E} = \{1, 2, \ldots, d\}$. The states are often called regimes in econometric literature. Given $\{R_t = s\}$, $s \in \mathbf{E}$, Y_t satisfies a GARCH equation with parameters $\alpha_i(s) \geq 0$, $1 \leq i \leq q$, $\beta_j(s) \geq 0$, $1 \leq j \leq p$ and $\omega(s) > 0$, which ensure that the conditional variance h_t is almost surely strictly positive. Suppose that $\{\eta_t\}$ and $\{R_t\}$ are independent and that the Markov chain is stationary, irreducible and aperiodic with stationary distribution $\pi(s) := P(R_1 = s)$, $1 \leq s \leq d$ and transition probabilities $p(k, l) := P(R_t = l | R_{t-1} = k)$. Also note that $\pi(s) > 0$ for all $s \in \mathbf{E}$ under assumptions. The Markov chain $\{R_t\}$ is unobservable and its transition probabilities and stationary distribution are unknown. Our aim is to draw statistical inference only based on the observed $\{Y_t\}$.

The parameters to be estimated from $\{Y_t\}$ are usually chosen to be

$$\boldsymbol{\theta} := \{p(k, l), \omega(s), \alpha_i(s), \beta_j(s), k \neq l, 1 \leq k, l, s \leq d, 1 \leq i \leq q, 1 \leq j \leq p\}.$$

The parameter space Θ is a compact subset of $\mathbb{R}^{d^2+pd+qd}$ and assume that the true parameter $\boldsymbol{\theta}_0 \in \Theta$. As Leroux [13, p. 135] has pointed out (see also Equation (9) below), asymptotically the stationary distribution $\pi(s)$, $1 \leq s \leq d$ will not affect the estimation. Here we assume p, q and d are known. See Rydén [14] and Francq *et al.* [4] for order selection problem.

From Equation (1) it follows that the conditional variance h_t depends on the present regime R_t, and the whole regime path as well, through h_{t-j}, $1 \leq j \leq p$. Because the number of possible regime paths grows exponentially with t, this recursive structure leads to an enormous terms in a likelihood as t increases, and the likelihood quickly becomes intractable. To avoid this path dependence problem, using ideas of Berkes *et al.* [12], we will transform the GARCH equation to an infinite order ARCH. Recall that the ARCH model is a special case of GARCH model with $p = 0$ in Equation (1) and the recursive structure disappears in the ARCH model.

First, we will give a sufficient and necessary condition for the existence of a strictly stationary solution for our model, following from Brandt [15], Bougerol and Picard [16] and Francq *et al.* ([4], Theorem 1). Let $r = \max(p, q)$ and by convention the $\alpha_i(R_t)$, $\beta_j(R_t)$ are equal to zero for $i > q$ and $j > p$.

Define

$$\boldsymbol{\tau}_t = (\beta_1(R_t) + \alpha_1(R_t)\eta_{t-1}^2, \ldots, \beta_{r-1}(R_t) + \alpha_{r-1}(R_t)\eta_{t-r+1}^2) \in \mathbb{R}^{r-1},$$

and the square matrix $\boldsymbol{A}_t$ of size r in block form as

$$\boldsymbol{A}_t = \begin{pmatrix} \boldsymbol{\tau}_t & \beta_r(R_t) + \alpha_r(R_t)\eta_{t-r}^2 \\ \boldsymbol{I}_{r-1} & \mathbf{0} \end{pmatrix},$$

where $\boldsymbol{I}_{r-1}$ is the identity matrix of size $r - 1$.

Let

$$\boldsymbol{B}_t = (\omega(R_t), 0, \ldots, 0)^T \in \mathbb{R}^r$$

and

$$\boldsymbol{X}_t = (h_t, h_{t-1}, \ldots, h_{t-r+1})^T \in \mathbb{R}^r.$$

Then Y_t is a solution of Equation (1) if and only if $\boldsymbol{X}_t$ is a solution of

$$\boldsymbol{X}_t = \boldsymbol{A}_t\boldsymbol{X}_{t-1} + \boldsymbol{B}_t, \quad t \in \mathbb{Z}. \tag{2}$$

Define the top Lyapunov exponent γ for $\{\boldsymbol{A}_t\}_{t\in\mathbb{Z}}$ as

$$\gamma = \inf_n \left\{ E\frac{1}{n} \log \| \boldsymbol{A}_n\boldsymbol{A}_{n-1} \ldots \boldsymbol{A}_1 \|, n \in \mathbb{N}^+ \right\}. \tag{3}$$

The following lemma is adapted from Francq *et al.* ([4], Theorem 1) (see also Brandt [15] and Bougerol and Picard [16]), which states that, roughly speaking, norm of $\boldsymbol{A}_t$ must be less than 1 'on average' to ensure a stationary and ergodic solution to Equation (2). The choice of norm is not very important since all norms in finite-dimensional vector spaces are equivalent; we can use, for instance, the Euclid norm.

LEMMA 2.1 *Equation* (2) *has a strictly stationary solution if and only if the top Lyapunov exponent* $\gamma < 0$. *Moreover, the stationary solution is unique, ergodic, and defined by*

$$\boldsymbol{X}_t = \boldsymbol{B}_t + \sum_{k=1}^{\infty} \boldsymbol{A}_t\boldsymbol{A}_{t-1} \ldots \boldsymbol{A}_{t-k+1}\boldsymbol{B}_{t-k}.$$

We are now in a position to give an ARCH representation to the GARCH equation. We need the following two assumptions:

(A1) η_1^2 is a non-degenerate random variable.
(A2) $E_{\boldsymbol{\theta}_0}[|\log p_\theta(Y_1)|] < \infty$, uniformly over $\boldsymbol{\theta} \in \Theta$.

We loosely use $p_{\boldsymbol{\theta}}(.)$ as marginal, joint or conditional probability density according to the context. (A1) is trivially satisfied if η_t is assigned a (regular) distribution. From Equation (1), (A2) implies that $E_{\boldsymbol{\theta}_0} \log h_1$ is finite. Define

$$\mathcal{A}_t(x) = \alpha_1(R_t)x + \alpha_2(R_t)x^2 + \cdots + \alpha_q(R_t)x^q$$

and

$$\mathcal{B}_t(x) = 1 - \beta_1(R_t)x - \beta_2(R_t)x^2 - \cdots - \beta_p(R_t)x^p.$$

Lemma 2.2 is adapted from Berkes *et al.* [12, Theorem 2.4].

LEMMA 2.2 *Assume* (A1) *and* (A2). *Then, for a strictly stationary and ergodic solution of* (1), *we have*

$$h_t = c_0(R_t) + \sum_{1 \le i < \infty} c_i(R_t) Y_{t-i}^2, \quad \forall\, t \in \mathbb{Z}$$

with probability one and this representation is unique with coefficients

$$c_0(R_t) = \frac{\omega(R_t)}{\mathcal{B}_t(1)}$$

and

$$c_n(R_t) = \frac{d^n}{dx^n}\left(\frac{\mathcal{A}_t(x)}{\mathcal{B}_t(x)}\right)_{x=0}, \quad 1 \le n < \infty.$$

Now, suppose that we are given a realization $\{y_1, \ldots, y_n\}$, while the Markov chain $\{R_t\}$ is not observed. We will use the maximum likelihood method for the estimation of regime-switching GARCH models. Assume that η_t is standard Gaussian. However, we conjecture that our results might be extended to more general distributional assumptions (like in Leroux [13]) and provide evidence in the simulation study (Section 4.2)[1]. From Equation (1), by summing up the probability density over all possible paths of the Markov chain, we get the likelihood function (r_t denotes the value of R_t):

$$p_{\boldsymbol{\theta}}(y_1, \ldots, y_n) = \sum_{(r_1, \ldots, r_n) \in \mathbf{E}^n} \pi(r_1) \left\{\prod_{t=2}^{n} p(r_{t-1}, r_t)\right\} \left\{\prod_{t=1}^{n} f_{r_t}(y_1, \ldots, y_t)\right\}, \tag{4}$$

given observations $y_1, \ldots, y_t$, where

$$f_{r_t}(y_1, \ldots, y_t) = \frac{1}{(2\pi h_{r_t}(y_1, \ldots, y_{t-1}))^{1/2}} \exp\left\{-\frac{y_t^2}{2h_{r_t}(y_1, \ldots, y_{t-1})}\right\},$$

and the conditional variance process follows

$$h_{r_t}(y_1, \ldots, y_{t-1}) = c_0(r_t) + \sum_{1 \le i \le t-1} c_i(r_t) y_{t-i}^2,$$

starting with

$$h_{r_1} = c_0(r_1), \quad h_{r_2}(y_1) = c_0(r_2) + c_1(r_2) y_1^2.$$

Recursive formulae for $\{c_i\}$ are available in Berkes *et al.* [12, pp. 210–211].

We can write Equation (4) as a product of matrices. It is possible because now the conditional variance and density at time t only depend on the current regime R_t, instead of the whole regime path, due to Lemma 2.1 and 2.2. Define vector $\mathbf{1} = (1, \ldots, 1)^T \in \mathbb{R}^d$, $\mathbf{p} = (\pi(1) f_1(y_1), \ldots, \pi(d) f_d(y_1))^T \in \mathbb{R}^d$ and matrix

$$\boldsymbol{M}_{\boldsymbol{\theta}}(y_1, \ldots, y_t) = \begin{pmatrix} p(1,1) f_1(y_1, \ldots, y_t) & p(2,1) f_1(y_1, \ldots, y_t) \ldots p(d,1) f_1(y_1, \ldots, y_t) \\ p(1,2) f_2(y_1, \ldots, y_t) & p(2,2) f_2(y_1, \ldots, y_t) \ldots p(d,2) f_2(y_1, \ldots, y_t) \\ \ldots & \ldots \\ p(1,d) f_d(y_1, \ldots, y_t) & p(2,d) f_d(y_1, \ldots, y_t) \ldots p(d,d) f_d(y_1, \ldots, y_t) \end{pmatrix},$$

then the likelihood is

$$p_{\boldsymbol{\theta}}(y_1, \ldots, y_n) = \mathbf{1}^T \left\{ \prod_{t=2}^{n} \boldsymbol{M}_{\boldsymbol{\theta}}(y_1, \ldots, y_t) \right\} \mathbf{p}, \tag{5}$$

which we can numerically maximize easily, using, for example, so-called forward–backward algorithm.

Define $p_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, Y_{t-2}, \ldots)$ as the density of Y_t given all previous observations until infinite past, $\{Y_{t-1}, Y_{t-2}, \ldots\}$, and $p_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, \ldots, Y_1)$ until Y_1. (Let $p_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, \ldots, Y_1) = p_{\boldsymbol{\theta}}(Y_1)$ when $t = 1$). Let $g_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, Y_{t-2}, \ldots)$ and $g_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, \ldots, Y_1)$ be the corresponding logarithms. The construction of $p_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, Y_{t-2}, \ldots)$ can be carried out as in Leroux [13]. In addition, under assumptions on the Markov chain and (A2), we can show that $E_{\boldsymbol{\theta}_0}[g_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, Y_{t-2}, \ldots)]$ and $E_{\boldsymbol{\theta}_0}[g_{\boldsymbol{\theta}}(Y_t | Y_{t-1}, \ldots, Y_1)]$ are finite, analogous to Leroux [13, Theorem 1 and 2]. Now, we define the conditional likelihood function based on all observations from infinite past, $\tilde{p}_{\boldsymbol{\theta}}(y_1, \ldots, y_n | y_0, y_{-1}, \ldots)$ similar to Equation (4), except replacing the density $f_{r_t}(y_1, \ldots, y_t)$ in Equation (4) by $p(y_t | y_{t-1}, y_{t-2}, \ldots; R_t = r_t)$. Furthermore, we can also write

$$\tilde{p}_{\boldsymbol{\theta}}(y_1, \ldots, y_n | y_0, y_{-1}, \ldots) = \mathbf{1}^T \left\{ \prod_{t=2}^{n} \boldsymbol{M}'_{\boldsymbol{\theta}}(y_t) \right\} \mathbf{p}', \tag{6}$$

where $\boldsymbol{M}'_{\boldsymbol{\theta}}(y_t)$ and $\mathbf{p}'$ replace $f_j(y_1, \ldots, y_t)$ by $p_{\boldsymbol{\theta}}(y_t | y_{t-1}, y_{t-2}, \ldots; R_t = j)$, $j = 1, \ldots, d$, in matrix $\boldsymbol{M}_{\boldsymbol{\theta}}(y_1, \ldots, y_t)$ and vector $\mathbf{p}$.

Because the indices of the states of the Markov chain can be permuted without changing the law of the model, the parameters are not strictly identifiable up to permutation. We introduce one identifiability condition.

(A3) *Identifiability Condition*: For any $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2 \in \Theta$ and all $Y_t, Y_{t-1}, \ldots$, if $p_{\boldsymbol{\theta}_1}(Y_t | Y_{t-1}, Y_{t-2}, \ldots) = p_{\boldsymbol{\theta}_2}(Y_t | Y_{t-1}, Y_{t-2}, \ldots)$, $P_{\boldsymbol{\theta}_0} - a.s.$, then $\boldsymbol{\theta}_1 = \boldsymbol{\theta}_2$.

3. The consistency of MLE

We will show the consistency of MLE for the regime-switching GARCH model (Theorem 3.1) here. Our method has benefited from the work of Francq *et al.* [4].

THEOREM 3.1 *Consider the regime-switching GARCH model* (1). *Assume that the top Lyapunov exponent* γ *defined in* (3) *is strictly negative. Assume also* (A2) *and* (A3). *Suppose that* Θ^* *is*

a compact subset of Θ *and* $\boldsymbol{\theta}_0 \in \Theta^*$; *For each* n, *let* $(\hat{\boldsymbol{\theta}}_n)$ *be an MLE sequence*, *with Gaussian innovations*, *satisfying almost surely*

$$p_{\hat{\boldsymbol{\theta}}_n}(Y_1, \ldots, Y_n) = \sup_{\boldsymbol{\theta} \in \Theta^*} p_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n),$$

then $(\hat{\boldsymbol{\theta}}_n)$ *tends almost surely to* $\boldsymbol{\theta}_0$ *as* $n \to \infty$.

Conditions describing the regime-switching GARCH model, including assumptions on the GARCH structure and (unobserved) Markov chain, are rather standard (see, e.g., [4, 9]). The strictly negative Lyapunov exponent γ ensures that the process $\{Y_t\}$, as well as $\{Y_t|Y_{t-1}, Y_{t-2} \ldots\}$ is strictly stationary and ergodic, which allows us to apply ergodic theorems. Berkes *et al.* [12] used $E_{\boldsymbol{\theta}_0} \log h_0 < \infty$, similar to (A2), to make sure that the transformation from GARCH to ARCH model is possible and unique. The existence of Θ^* is trivial. In fact, we can enlarge Θ^* if necessary, even to include $\boldsymbol{\theta}_0$ as an interior of Θ^*.

We will prove Lemma 3.2, 3.3 and 3.4. Theorem 3.1 easily follows frow Lemma 3.4 together with the identifiability condition. The following lemma 3.2 shows that asymptotically, the likelihood (4) is equivalent to the conditional likelihood $\tilde{p}_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n|Y_0, Y_{-1}, \ldots)$ for all $\boldsymbol{\theta} \in \Theta$.

LEMMA 3.2 *Consider model* (1). *Assume the top Lyapunov exponent defined in* (3) *strictly negative and* (A2). *For all* $\boldsymbol{\theta} \in \Theta$,

$$\begin{aligned} \lim_{n\to\infty} \frac{1}{n} \log p_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n) &= \lim_{n\to\infty} \frac{1}{n} \log \tilde{p}_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n|Y_0, Y_{-1}, \ldots) \\ &= E_{\boldsymbol{\theta}_0} \log g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, \ldots). \end{aligned} \tag{7}$$

Proof First, note that

$$\log \tilde{p}_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n|Y_0, Y_{-1}, \ldots) = \sum_{t=1}^{n} g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, Y_{t-2}, \ldots)$$

and

$$\log p_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n) = \sum_{t=1}^{n} g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, \ldots, Y_1).$$

Write

$$\begin{aligned} \frac{1}{n} \sum_{t=1}^{n} g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, \ldots, Y_1) &= \frac{1}{n} \sum_{t=1}^{n} g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, Y_{t-2}, \ldots) + \frac{1}{n} \sum_{t=1}^{n} \{g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, \ldots, Y_1) \\ &\quad - g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, Y_{t-2}, \ldots)\}. \end{aligned} \tag{8}$$

Analogous to Karlin and Taylor [17, pp. 502], define

$$\phi_N(y_0, y_{-1}, \ldots) = \sup_{l \geq N} |g_{\boldsymbol{\theta}}(y_0|y_{-1}, \ldots, y_{-l}) - g_{\boldsymbol{\theta}}(y_0|y_{-1}, y_{-2}, \ldots)|,$$

and

$$Z_t^N = \phi_N(Y_t, Y_{t-1}, \ldots).$$

Then $\{Z_t^N\}$ is stationary, ergodic, and $E[|Z_t^N|] < \infty$ under assumptions. We have

$$\begin{aligned}
&\limsup_{n\to\infty} \left| \frac{1}{n} \sum_{t=1}^{n} \{g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, \ldots, Y_1) - g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, Y_{t-2}, \ldots)\} \right| \\
&\quad \leq \limsup_{n\to\infty} \frac{1}{n} \sum_{t=1}^{n} |g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, \ldots, Y_1) - g_{\boldsymbol{\theta}}(Y_t|Y_{t-1}, Y_{t-2}, \ldots)| \\
&\quad \leq \limsup_{n\to\infty} \frac{1}{n} \sum_{t=N+1}^{n} Z_t^N = E[Z_1^N].
\end{aligned}$$

But as $N \to \infty$, $Z_1^N \to 0$, and the interchange of limit and expectation can be justified to give $\lim_{N\to\infty} E[Z_1^N] = 0$. So the second term on the right-hand side of (8) goes to zero as $n \to \infty$. The second equality in Equation (7) follows by applying ergodic theorem (e.g. [18, Corollary 3.5.1]), which completes the proof. ■

We will next compare the likelihood $p_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n)$ with the one evaluated at the true parameter $\boldsymbol{\theta}_0$, $p_{\boldsymbol{\theta}_0}(Y_1, \ldots, Y_n)$. Define

$$M_n(\boldsymbol{\theta}) = \frac{1}{n} \log \frac{p_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n)}{p_{\boldsymbol{\theta}_0}(Y_1, \ldots, Y_n)},$$

and the following lemma follows from Lemma 3.2, Jensen's inequality and identifiability assumption.

LEMMA 3.3 *Under the conditions of Lemma* 3.2, *for all* $\boldsymbol{\theta} \in \Theta$, *with probability one,*

$$\lim_{n\to\infty} \frac{1}{n} \log \frac{\tilde{p}_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n|Y_0, Y_{-1}, \ldots)}{\tilde{p}_{\boldsymbol{\theta}_0}(Y_1, \ldots, Y_n|Y_0, Y_{-1}, \ldots)} = \lim_{n\to\infty} M_n(\boldsymbol{\theta}) \leq 0$$

and the limit is almost surely equal to zero if and only if $\boldsymbol{\theta} = \boldsymbol{\theta}_0$.

LEMMA 3.4 *Under the conditions of Lemma* 3.2, *for any* $\boldsymbol{\theta}_1 \in \Theta$, $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_0$, *there exists a neighborhood* $V(\boldsymbol{\theta}_1)$ *of* $\boldsymbol{\theta}_1$ *such that*

$$\limsup_{n\to\infty} \sup_{\boldsymbol{\theta}\in V(\boldsymbol{\theta}_1)} M_n(\boldsymbol{\theta}) < 0 \text{ a.s.}$$

Proof First, define the matrix norm $\|\cdot\|$ as the sum of all element of the matrix. Making use of the product of matrices (6), we have

$$\begin{aligned}
&\min_j \pi(j) p(Y_1|Y_0, Y_{-1}, \ldots; R_t = j) \left\| \prod_{t=2}^{n} M'_{\boldsymbol{\theta}}(Y_t) \right\| \leq \tilde{p}_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n|Y_0, Y_{-1}, \ldots) \\
&= \left\| \left\{ \prod_{t=2}^{n} M'_{\boldsymbol{\theta}}(Y_t) \right\} \mathbf{p}' \right\| \leq \max_j \pi(j) p(Y_1|Y_0, Y_{-1}, \ldots; R_t = j) \left\| \prod_{t=2}^{n} M'_{\boldsymbol{\theta}}(Y_t) \right\|.
\end{aligned}$$

Hence, from assumption (A2) it is straightforward to obtain

$$\lim_{n\to\infty} \frac{1}{n} \log \tilde{p}_{\boldsymbol{\theta}}(Y_1, \ldots, Y_n|Y_0, Y_{-1}, \ldots) = \lim_{n\to\infty} \frac{1}{n} \log \left\| \prod_{t=2}^{n} M'_{\boldsymbol{\theta}}(Y_t) \right\|. \tag{9}$$

Next, let $V_r(\boldsymbol{\theta}_1)$ be the open sphere with centre $\boldsymbol{\theta}_1$ and radius $1/r$ and define

$$S^r_{2,n}(Y_n) = \sup_{\boldsymbol{\theta}\in V_r(\boldsymbol{\theta}_1)} \left\| \prod_{t=2}^{n} M'_{\boldsymbol{\theta}}(Y_t) \right\|.$$

Because this matrix norm is multiplicative, we have, for $\boldsymbol{\theta} \in V_r(\boldsymbol{\theta}_1)$

$$\sup_{\boldsymbol{\theta}} \left\| \prod_{t=2}^{n+k} M'_{\boldsymbol{\theta}}(Y_t) \right\| \le \sup_{\boldsymbol{\theta}} \left\| \prod_{t=2}^{n} M'_{\boldsymbol{\theta}}(Y_t) \right\| \cdot \sup_{\boldsymbol{\theta}} \left\| \prod_{t=n+1}^{n+k} M'_{\boldsymbol{\theta}}(Y_t) \right\|,$$

that is

$$\log S^r_{2,n+k}(Y_{n+k}) \le \log S^r_{2,n}(Y_n) + \log S^r_{n+1,n+k}(Y_{n+k})$$

for any positive integers $n(\ge 2)$, k and r. From the Kingman [19] ergodic theorem for the subadditive processes we can readily get

$$\lim_{n\to\infty} \frac{1}{n} \log S^r_{2,n}(Y_n) = \lambda_r(\boldsymbol{\theta}_1) := \inf_{n>1} \frac{1}{n} E_{\boldsymbol{\theta}_0} \log S^r_{2,n}(Y_n),\ P_{\boldsymbol{\theta}_0} - a.s.$$

Recall that almost surely

$$\lambda(\boldsymbol{\theta}_0) := \lim_{n\to\infty} \frac{1}{n} \log \left\| \prod_{t=2}^{n} M'_{\boldsymbol{\theta}_0}(Y_t) \right\| = \inf_{n>1} \frac{1}{n} E_{\boldsymbol{\theta}_0} \log \left\| \prod_{t=2}^{n} M'_{\boldsymbol{\theta}_0}(Y_t) \right\|.$$

Thus, from Lemma 3.3 and Equation (9), there exist $\varepsilon > 0$ and $n_\varepsilon \in \mathbb{N}$ such that

$$\frac{1}{n_\varepsilon} E_{\boldsymbol{\theta}_0} \log \left\| \prod_{t=2}^{n_\varepsilon} M'_{\boldsymbol{\theta}_1}(Y_t) \right\| < \lambda(\boldsymbol{\theta}_0) - \varepsilon.$$

The dominated convergence theorem shows that, for r large enough,

$$\lambda_r(\boldsymbol{\theta}_1) \le \frac{1}{n_\varepsilon} E_{\boldsymbol{\theta}_0} \log S^r_{2,n_\varepsilon}(Y_{n_\varepsilon}) < \lambda(\boldsymbol{\theta}_0) - \frac{\varepsilon}{2}.$$

Applying Lemma 3.2 again completes the proof. ■

4. Simulation studies

In our simulation studies, we will illustrate the consistency, investigate asymptotic normality of both Gaussian and non-Gaussian MLE, and analyse a model specification problem. The most commonly used two-regime switching model is considered.

Given the recursive form of likelihood (5), we can numerically maximize it, following the idea of MacDonald and Zucchini ([9, pp. 78–79]). During the estimation, besides the standard constraints $0 \le p(k,l) \le 1$, $\sum_{k\ne l} p(k,l) \le 1$, $\alpha_i(s) \ge 0$, $\beta_j(s) \ge 0$, and $\sum_{i=1}^{q} \alpha_i(s) + \sum_{j=1}^{p} \beta_j(s) \le 1$, we also impose $\omega(s) \ge 0.001$ to avoid the underflow of numerical algorithm, and $\omega(1) \le \omega(2) \le \cdots \le \omega(d)$ for identifiability. Several randomly chosen starting point sets are used to pursue the global maxima.

Table 1. True values and the mean and standard deviation of MLE of regime-switching GARCH with Gaussian innovations.

	$\hat{\omega}(1)$	$\hat{\alpha}_1(1)$	$\hat{\beta}_1(1)$	$\hat{\omega}(2)$	$\hat{\alpha}_1(2)$	$\hat{\beta}_1(2)$	$\hat{p}(1,2)$	$\hat{p}(2,1)$
True	1	0.4	0.2	20	0.2	0.4	0.1	0.11
mean	1.015	0.419	0.213	17.112	0.219	0.424	0.096	0.093
sd.	0.11	0.058	0.039	3.13	0.038	0.085	0.009	0.011

4.1. *Asymptotics for Gaussian MLE*

In the first experiment to verify the consistency of MLE, we generate 100 independent series from model (1), each with size 5000. The true model has two regimes. The corresponding parameters are same as in Xie and Yu [11]: $\omega(1) = 1$, $\alpha_1(1) = 0.4$, $\beta_1(1) = 0.2$ and $\omega(2) = 20$, $\alpha_1(2) = 0.2$, $\beta_1(2) = 0.4$. The transition probabilities are set to $p(1, 2) = 0.1$ and $p(2, 1) = 0.1$. These parameters are chosen such that they are fairly away from the boundary of parameter space. Table 1 summarizes the true values and the mean and standard deviation of estimators.

Although the consistency of MLE of our model is fairly satisfying from Table 1, we should bear in mind that we need distributional property of the estimates to utilize the information of standard deviations. With the second experiment, we want to examine the asymptotic normality through simulation. Bickel *et al.* [20] proved the asymptotic normality of MLE for general HMM model. However, their results cannot be readily extended to the regime-switching GARCH model. Xie and Yu [11] conjectured that the asymptotic distribution of quasi-MLE is normal for the reduced regime-switching GARCH model by simulation study.

First, note that by taking away the GARCH effect, i.e., letting $p = 0$ in model (1), our model is the same as the regime-switching ARCH model considered by Francq *et al.* [4]. Douc *et al.* [21] showed the normality of the regime-switching ARCH model in a more general autoregressive model, making use of the uniform exponential forgetting the initial distribution for the hidden Markov chain conditional on the observations (see [21] for details). Therefore, we will only investigate the asymptotic normality of regime-switching GARCH model here. The experiment is the same as the first one. The QQ-plots with 0-1 line are shown in Figure 1. They rather conform to the family of normal distributions. The P-values of the Kolmogorov–Smirnov goodness-of-fit test verify our observation from QQ-plots. They are all above 0.1,[2] and the hypothesis of normal distributions cannot be rejected in a reasonable level. We can still expect better conformity to normal distributions if we increase the length of sequences to, for example, 10,000, from our experience of the improvement already achieved by increasing the length from 2000 to 5000. We compromised for the vast time-consumption on computation and gave up trying.

4.2. *Asymptotics for non-Gaussian MLE*

We see no essential difficulty to extend the consistency result to non-Gaussian MLE, or even to a general density with suitable assumptions such as those in Leroux [13]. But we keep the simple structure in the theoretical proof and illustrate the consistency, as well as the asymptotic normality of non-Gaussian MLE, by numerical examples using two-component mixture normal distributions. A two-component mixture normal distribution has the density $(1 - p)\phi(\mu_1, \sigma_1^2) + p\phi(\mu_2, \sigma_2^2)$, where $\phi(\mu, \sigma^2)$ is an univariate Gaussian density with mean μ and variance σ^2 and p is the mixing probability. With small p, this distribution is sometimes called contaminated normal and can be used to examine the departure of sample observations from a normal distribution. Rachev and Mittnik [23, Ch.4] compared the likelihood and goodness-of-fit of several distributions modelled to financial return data and the mixture normal was shown competent (although usually

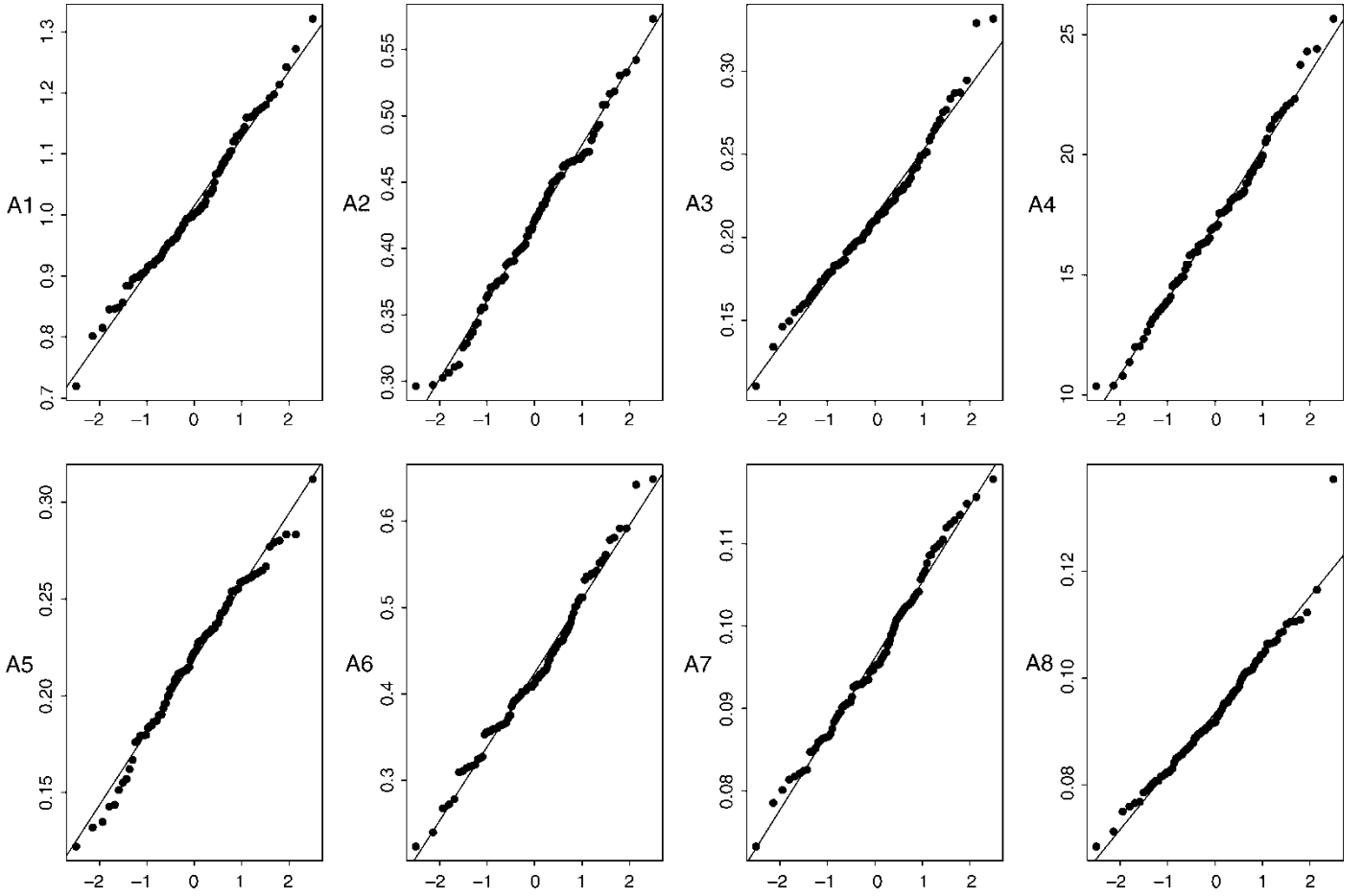


Figure 1. The QQ-plot of MLE for regime-switching GARCH model with 0-1 line: A1–A8 represent estimators of $\omega(1)$, $\omega(2)$, $\alpha_1(1)$, $\alpha_1(2)$, $\beta_1(1)$, $\beta_1(2)$, $p(1, 2)$ and $p(2, 1)$, respectively. See text in section 4.1 for detail.

Table 2. The standard deviation and mean of bias (in parentheses) of MLE for regime-switching GARCH model with mixture normal innovations, for different sample sizes n.

	$\hat{\omega}(1)$	$\hat{\alpha}_1(1)$	$\hat{\beta}_1(1)$	$\hat{\omega}(2)$	$\hat{\alpha}_1(2)$	$\hat{\beta}_1(2)$	$\hat{p}(1,2)$	$\hat{p}(2,1)$
$n = 500$	0.315	0.230	0.127	12.24	0.186	0.225	0.042	0.037
	(0.011)	(0.064)	(0.031)	(6.83)	(0.138)	(−0.012)	(0.014)	(−0.005)
$n = 2000$	0.148	0.131	0.074	6.839	0.126	0.134	0.015	0.014
	(−0.01)	(0.01)	(0.042)	(5.538)	(0.155)	(0.004)	(0.016)	(−0.001)
$n = 5000$	0.091	0.082	0.045	4.984	0.064	0.105	0.011	0.012
	(−0.004)	(0.008)	(0.035)	(4.849)	(0.132)	(0.016)	(0.013)	(−0.004)

not better than the stable Paretian distributions studied in that monogaph). This density can have flexible shapes with different parameters.

For illustrative purpose, we report the results for a relatively extreme bimodal mixture normal with $\mu_1 = 0.25, \sigma_1^2 = 0.3, \mu_2 = -2.25, \sigma_2^2 = 1.675$ and $p = 0.1$. It has zero mean, unit variance: skewness around -2.05 and kurtosis 8.84. For other unimodal mixtures which are closer to normal distributions, the results obtained are similar to that we report. For this innovation density, samples of sizes 500, 2000 and 5000, respectively, each with 50 replications are generated from (1) with same parameters as in the first experiment. The MLE are obtained by maximizing the likelihood (5) with the conditional density $f_{r_1}(\cdot)$ in (4) replaced by the corresponding mixture normal. The standard deviations (and means of bias) of these MLE for different sample sizes are summarized in Table 2. We can see that while the biases generally decrease, the standard deviations always decrease as the sample size increases, which suggests the estimates are consistent. Moreover, the asymptotic normality of the MLE is also supported. For the estimates with sample size 5000, the QQ-points (not shown here) are well close to the reference 0-1 line and the P-values of 1ts.gof test are all above 0.1 except that for parameter $p(1, 2)$, which is 0.072. Again, using larger numbers of replications and/or sample sizes may improve the result, but we stop trying considering the vast time-consuming in the resulted computation.

4.3. *A model specification problem*

In this experiment, we are interested in the consequence if we wrongly specify an ordinary GARCH to a regime-switching process; we mainly focus us on the persistence of model, i.e., how close to 1 the sum of estimated parameters of the wrongly specified model, $\sum_{i=1}^{q} \hat{\alpha}_i + \sum_{j=1}^{p} \hat{\beta}_j$, will be. Since the strong persistence of the ordinary GARCH model can be possibly explained by regime-switching models, it leads us to guess that, on the reverse, using ordinary GARCH model to fit data generated from regime-switching models will generally result in strong persistence. Previous studies (see, e.g., ref. [3]) did give us such an impression. However, we argue here that this is not generally true and actually depends on the transition probabilities between different regimes.

Table 3 gives the estimates when data come from the two regimes ARCH model. The true parameters have been used by Francq *et al.* [4]: $\omega(1) = 1, \alpha_1(1) = 0.6, \omega(2) = 100, \alpha_1(2) = 0$, $p(1, 2) = p(2, 1) = 0.01$. Then we change pair of $p(1, 2)$ and $p(2, 1)$ to (0.1, 0.1), (0.2, 0.2) and (0.5, 0.5). We generate 100 independent series with size 1000 for each parameter set and apply them to the ordinary GARCH(1, 1) model. The estimates for ω and sum of GARCH parameters, $\alpha + \beta$ are summarized. Our conclusion is quite similar to that of Xie and Yu [11] for the reduced regime-switching model. We see that while $p(1, 2)$ and $p(2, 1)$ are small, $\hat{\alpha} + \hat{\beta}$ is close to 1, which shows strong persistence and is sometimes said to be 'close to non-stationary region' in the sense of the weak stationarity of Bollerslev [6]. This agrees with our previous knowledge. However, it is interesting that as $p(1, 2)$ and $p(2, 1)$ increase, the estimates move from the non-stationary region to approximately the average of these two regimes, which is far from persistent or

Table 3. Estimate result fitting an ordinary GARCH(1, 1) model to data generated from two-regime ARCH models with different transition probabilities, where $\hat{\omega}$, $\hat{\alpha}$ and $\hat{\beta}$ are estimates of parameters of the ordinary GARCH model.

$(p(1,2)\ p(2,1))$	$(0.01, 0.01)$	$(0.1, 0.1)$	$(0.2, 0.2)$	$(0.5, 0.5)$
$\hat{\omega}$	0.6695	8.5987	22.3172	41.0445
$\hat{\alpha} + \hat{\beta}$	0.9994	0.9587	0.7046	0.4205

Table 4. Estimate result fitting an ordinary GARCH(1, 1) model to data generated from two-regime GARCH models with different transition probabilities.

$(p(1,2)\ p(2,1))$	$(0.01, 0.01)$	$(0.1, 0.1)$	$(0.2, 0.2)$	$(0.5, 0.5)$
$\hat{\omega}$	0.3192	2.621	5.8904	10.0464
$\hat{\alpha} + \hat{\beta}$	0.9918	0.8683	0.6498	0.429

non-stationary. Table 4 gives us a similar result when the data are from the two-regime switching GARCH model used in Section 4.1.

5. Final remarks

The regime-switching GARCH model, along the lines of HMM, incorporates the idea of regime-switching into the more restrictive GARCH model. This incorporation is desirable and significantly extends the GARCH model. In fact, not only the parameters in the GARCH equation but also the structure of model and error distribution can be specified to depend on regimes, c.f. Gray [3], Hamilton and Raj [24] and Douc *et al.* [21]. There is one difficult problem with this extension, that is, observations will then depend on the whole regime path because of the recursive structure of the GARCH equation. It causes an intractable likelihood and limits the usefulness of the general regime-switching GARCH model. Only the regime-switching ARCH (e.g., ref. [1,2]) or some kind of reduced GARCH model [3,10] is feasible to use in practice. In this paper, following the transform proposed by Berkes *et al.* [12], we rewrote the GARCH model as ARCH(∞), which overcame the regime path-dependence problem and led to a likelihood that can be handled directly. It makes the estimation and empirical applications of the general regime-switching GARCH model possible. The consistency of the MLE was also proved.

Asymptotics normality is another issue we are interested in. Bickel *et al.* [20] proved the asymptotic normality for general HMM, but the conditional independence of observations given the unobserved regime is crucial in their framework and is not easy to be relaxed to include our model. Douc *et al.* [21] obtained the asymptotic normality for a class of autoregressive models with Markov regime, which includes HMM and regime-switching ARCH model as special cases. However, they instantly use the Markov property of $\{R_k, Y_k, \ldots, Y_{k-s+1}\}$ (assume an s-th order ARCH model), which doesn't hold for GARCH models. Our numerical experiment supported the hypothesis that the MLE for regime-switching GARCH model is asymptotically normal, but we need to find approaches, other than what we have used to prove the consistency, to theoretically prove it. It will be a challenging, but rewarding task in the future.

As for the model specification problem, we observed that if the two regimes of the true process seldom switch to each other, i.e., the transition probabilities between these regimes are fairly small, wrongly specifying an ordinary GARCH model to such a process will generally result in strong persistence or non-stationarity. However, if the regimes switch quite often, it is difficult to detect the wrong specification just in terms of persistence or weak stationarity. This implies

that even a usual stationary GARCH process may actually be generated from a regime-switching GARCH model (with large transition probabilities between different regimes), which we haven't realized before and may lead to interesting applications.

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Notes

1. The author thanks an anonymous referee for the insightful comments on this issue.
2. The P-value is calculated using the approximation of Dallal and Wilkinson [22], which is most accurate for P-value less than 0.1. So these P-values are all set to 0.5 in statistical software S-plus®, with which we carry out our analysis.

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